Bài Scaling

$d_{k_2} = 4$

$QK^T = [4 \quad 2 \quad 0]$

$\frac{QK^T}{\sqrt{4}} = [2 \quad 1 \quad 0]$

Softmax : $\frac{QK^T}{\sqrt{4}}$

$\alpha_1 = \frac{e^2}{e^2 + e^1 + e^0} = 0,6652$

$\alpha_2 = \frac{e^1}{e^2 + e^1 + e^0} = 0,2447$

$\alpha_3 = \frac{e^0}{e^2 + e^1 + e^0} = 0,09$

Softmax : QK^T

$\alpha_1 = 0,8668$

$\alpha_2 = 0,1173$

$\alpha_3 = 0,0158$

Bài 4

$q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$k_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, k_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, k_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$s_i = q^T \cdot k_i$

$s_1 = 1, s_2 = 0, s_3 = 1$

$S = [1, 0, 1]$

Softmax

$\alpha_i = \frac{e^{s_i}}{\sum_{j=1}^{3} e^{s_j}}$

$\alpha_1 = \frac{e^1}{e^1 + e^0 + e^1} = 0,422$

$\alpha_2 = \frac{e^0}{e^1 + e^0 + e^1} = 0,155$

$\alpha_3 = \frac{e^1}{e^1 + e^0 + e^1} = 0,422$

$\Rightarrow \alpha \approx [0,422, 0,155, 0,422]$

Lab 6

Bài 1: $q = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$

$s_1 = q^T \cdot k_1$

$= [1 \ 2] \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$

$s_2 = q^T \cdot k_2$

$= [1 \ 2] \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2$

$s_3 = q^T \cdot k_3$

$= [1 \ 2] \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3$

Bài 2: $s = [1, 2, 3]$

$\alpha = \text{softmax}(s)$

$= \dfrac{e^{s_i}}{\sum_j e^{s_j}}$ = ~~[illegible]~~

~~$\alpha_1 = e^1 = 2,718$~~

~~$\alpha_2 = e^2 = 7,389$~~

~~$\alpha_3 = e^3 = 20,08$~~

$\alpha_1 = \dfrac{e^1}{e^1+e^2+e^3} = 0,09$

$\alpha_2 = \dfrac{e^2}{e^1+e^2+e^3} = 0,245$

$\alpha_3 = \dfrac{e^3}{e^1+e^2+e^3} = 0,665$

Bài 3:

$\alpha = [0,1, 0,2, 0,7]$

$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\quad v_3 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

v_3 có trọng số 0.7 đóng góp nhiều nhất

$c = 0,1 v_1 + 0,2 v_2 + 0,7 v_3$

$= \begin{bmatrix} 0,1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0,2 \end{bmatrix} + \begin{bmatrix} 1,4 \\ 1,4 \end{bmatrix}$

$= \begin{bmatrix} 1,5 \\ 1,6 \end{bmatrix}$